Class 09: Candy Mini-Project

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Background

This analysis will explore a Halloween Candy Dataset by FiveThirtyEight. The goal is to analyze and interpret: **What is the top ranked snack-sized candy?**. What makes this candy more desirable: price? sugar? chocolate?

Importing the Candy Data Set

This is importing the Candy Data Set

```
candy_file <- "candy-data.csv"
candy <- read.csv(candy_file, row.names = 1)
head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
```

```
One dime              0       0       0                 0       0                 0
One quarter           0       0       0                 0       0                 0
Air Heads             0       1       0                 0       0                 0
Almond Joy            1       0       0                 1       0                 0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
3 Musketeers    0   1        0        0.604        0.511   67.60294
One dime        0   0        0        0.011        0.116   32.26109
One quarter     0   0        0        0.011        0.511   46.11650
Air Heads       0   0        0        0.906        0.511   52.34146
Almond Joy      0   1        0        0.465        0.767   50.34755
```

> Q1. How many different candy types are in this dataset?

85 different types.

```r
nrow(candy)
```

```
[1] 85
```

> Q2. How many fruity candy types are in the dataset?

There are 38 types of fruity candy.

```r
sum(candy$fruity)
```

```
[1] 38
```

> Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

My favorite candy is reese's peanut butter cups.

```r
candy["Reese's Peanut Butter cup", ]$winpercent
```

```
[1] 84.18029
```

> Q4. What is the winpercent value for "Kit Kat"?

The winpercent value for kit kat candy is 76.7686.

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

> Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

The winpercent value for tootsie rolls is 49.6535.

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

An alternative way to find the preference for a candy over another randomly chosen is with `dplyr`. Loading `dplyr` using **library(dplyr)** will bring this into R.

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
     winpercent
Twix   81.64291
```

There is a useful `skim()` function in the **skimr** package that can help give you a quick overview of a given dataset. Let's install this package try it on our candy data.

```r
library(skimr)
```

```r
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

> Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Yes: the column winpercent is on a different scale compared to the majority of other columns in the dataset.

> Q7. What do you think a zero and one represent for the candy$chocolate column?

The candy$chocolate column is a binary variable showing presence or absence of chocolate.

Exploratory Analysis

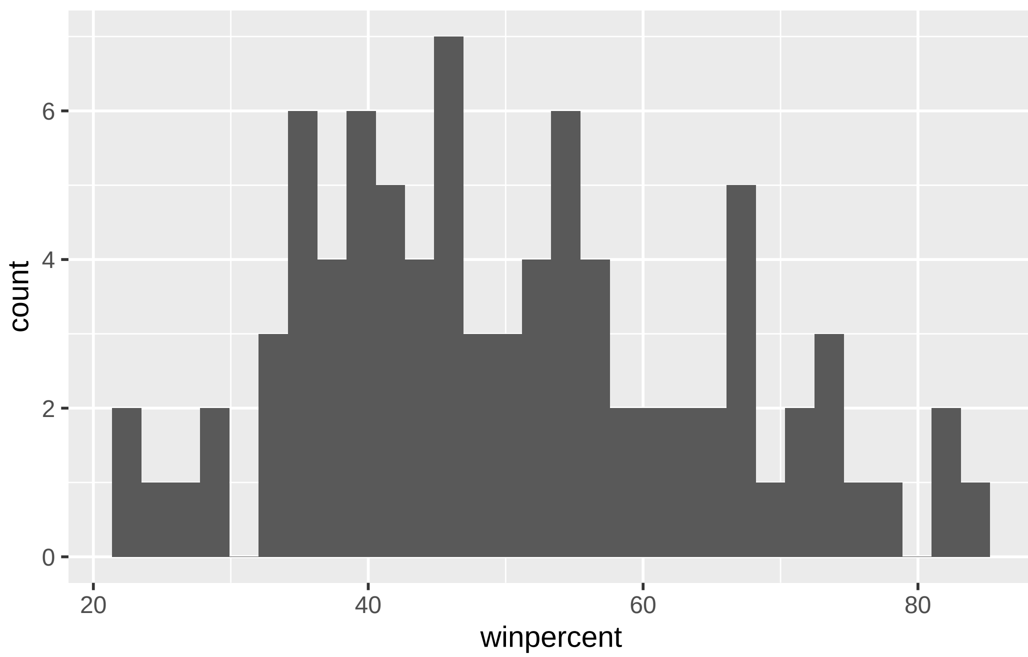
A good place to start for exploratory analysis is with a histogram. The base R function `hist()` is perfect for this, although more complex plots with `ggplot()` using `geom_hist()` can also be achieved.

> Q8. Plot a histogram of winpercent values using both base R and ggplot2=

```r
library(ggplot2)

ggplot(candy) +
  aes(winpercent) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



> Q9: Is the distribution of winpercent values symmetrical?

The distribution isn't symmetrical as measured by the center; there is a slight right skew. We can use **median** to measure centrality because it represents the "middle" of the data.

```
summary(candy$winpercent)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  22.45   39.14   47.83   50.32   59.86   84.18 
```

```
median(candy$winpercent)
```

```
[1] 47.82975
```

> Q10. Is the center of the distribution above or below 50%?

The center of the distribution is below 50%.

```
summary(candy$winpercent)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  22.45   39.14   47.83   50.32   59.86   84.18 
```

> Q11. On average is chocolate candy higher or lower ranked than fruit candy?

Chocolate candies are ranked higher on average than fruity candies:

```
choc.ind <- as.logical(candy$chocolate)
choc.candy <- candy[choc.ind,]
choc.win <- choc.candy$winpercent
mean(choc.win)
```

```
[1] 60.92153
```

```
fruit.ind <- as.logical(candy$fruity)
fruit.candy <- candy[fruit.ind,]
fruit.win <- fruit.candy$winpercent
mean(fruit.win)
```

```
[1] 44.11974
```

> Q12. Is this difference statistically significant?

Yes, the difference is statistically significant ($p < 0.05$), see below:

```r
t.test(choc.win, fruit.win)
```

```
	Welch Two Sample t-test

data:  choc.win and fruit.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y 
 60.92153  44.11974 
```

Overall Candy Rankings

> Q13. What are the five least liked candy types in this set?

The five least liked candy types in this data set are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

```r
head(candy$winpercent)
```

```
[1] 66.97173 67.60294 32.26109 46.11650 52.34146 50.34755
```

```r
order(candy$winpercent)
```

```
 [1] 45  8 13 73 27 58 72  3 71 20 10 70 60 56 12 51 49 63  9 11 82 31 17 46 15
[26] 50 30 84 22 14 59 76 16 83 81 77 64  4 47 35 18 79 40 75 85 78  6 21  5 68
[51] 32 41 74 36 62 42 23 25  7 19 28 26 66 67 38 24 61 39 57 44 34  1 69  2 48
[76] 43 33 55 37 54 65 29 80 52 53
```

```r
candy[order(candy$winpercent), ][1:5, ]
```

```
                   chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                  0      1       0              0      0
Boston Baked Beans         0      0       0              1      0
Chiclets                   0      1       0              0      0
Super Bubble               0      1       0              0      0
```

```
Jawbusters                        0       1        0                0       0
                    crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                          0    0   0        1        0.197        0.976
Boston Baked Beans                 0    0   0        1        0.313        0.511
Chiclets                           0    0   0        1        0.046        0.325
Super Bubble                       0    0   0        0        0.162        0.116
Jawbusters                         0    1   0        1        0.093        0.511
                    winpercent
Nik L Nip             22.44534
Boston Baked Beans    23.41782
Chiclets              24.52499
Super Bubble          27.30386
Jawbusters            28.12744
```

Q14. What are the top 5 all time favorite candy types out of this set?

The five favorite candy types in this data set are Reese's Peanut Butter cups, Reese's Miniatures, Twix, Kit Kat, and Snickers.

```
head(candy$winpercent)
```

```
[1] 66.97173 67.60294 32.26109 46.11650 52.34146 50.34755
```

```
order(candy$winpercent)
```

```
 [1] 45  8 13 73 27 58 72  3 71 20 10 70 60 56 12 51 49 63  9 11 82 31 17 46 15
[26] 50 30 84 22 14 59 76 16 83 81 77 64  4 47 35 18 79 40 75 85 78  6 21  5 68
[51] 32 41 74 36 62 42 23 25  7 19 28 26 66 67 38 24 61 39 57 44 34  1 69  2 48
[76] 43 33 55 37 54 65 29 80 52 53
```

```
candy[order(-candy$winpercent), ][1:5, ]
```

```
                          chocolate fruity caramel peanutyalmondy nougat
Reese's Peanut Butter cup         1      0       0              1      0
Reese's Miniatures                1      0       0              1      0
Twix                              1      0       1              0      0
Kit Kat                           1      0       0              0      0
Snickers                          1      0       1              1      1
                          crispedricewafer hard bar pluribus sugarpercent
Reese's Peanut Butter cup                0    0   0        0        0.720
Reese's Miniatures                       0    0   0        0        0.034
```

```
Twix                                     1     0    1         0         0.546
Kit Kat                                  1     0    1         0         0.313
Snickers                                 0     0    1         0         0.546
                          pricepercent winpercent
Reese's Peanut Butter cup        0.651   84.18029
Reese's Miniatures               0.279   81.86626
Twix                             0.906   81.64291
Kit Kat                          0.511   76.76860
Snickers                         0.651   76.67378
```

```r
x <- c(5,10,1)
sort(x)
```

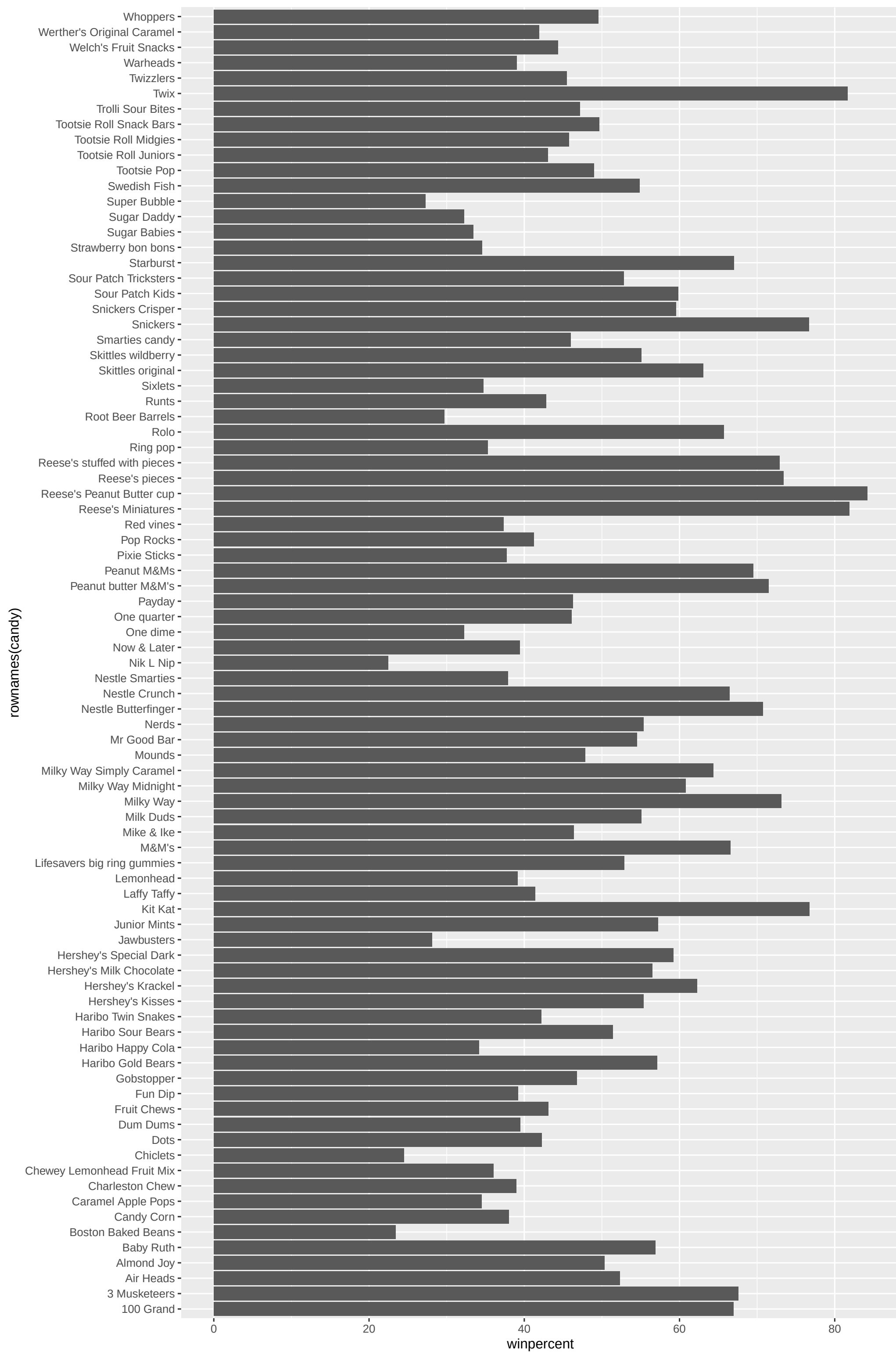
```
[1]  1  5 10
```

```r
order(x)
```

```
[1] 3 1 2
```

> Q15. Make a first barplot of candy ranking based on winpercent values.

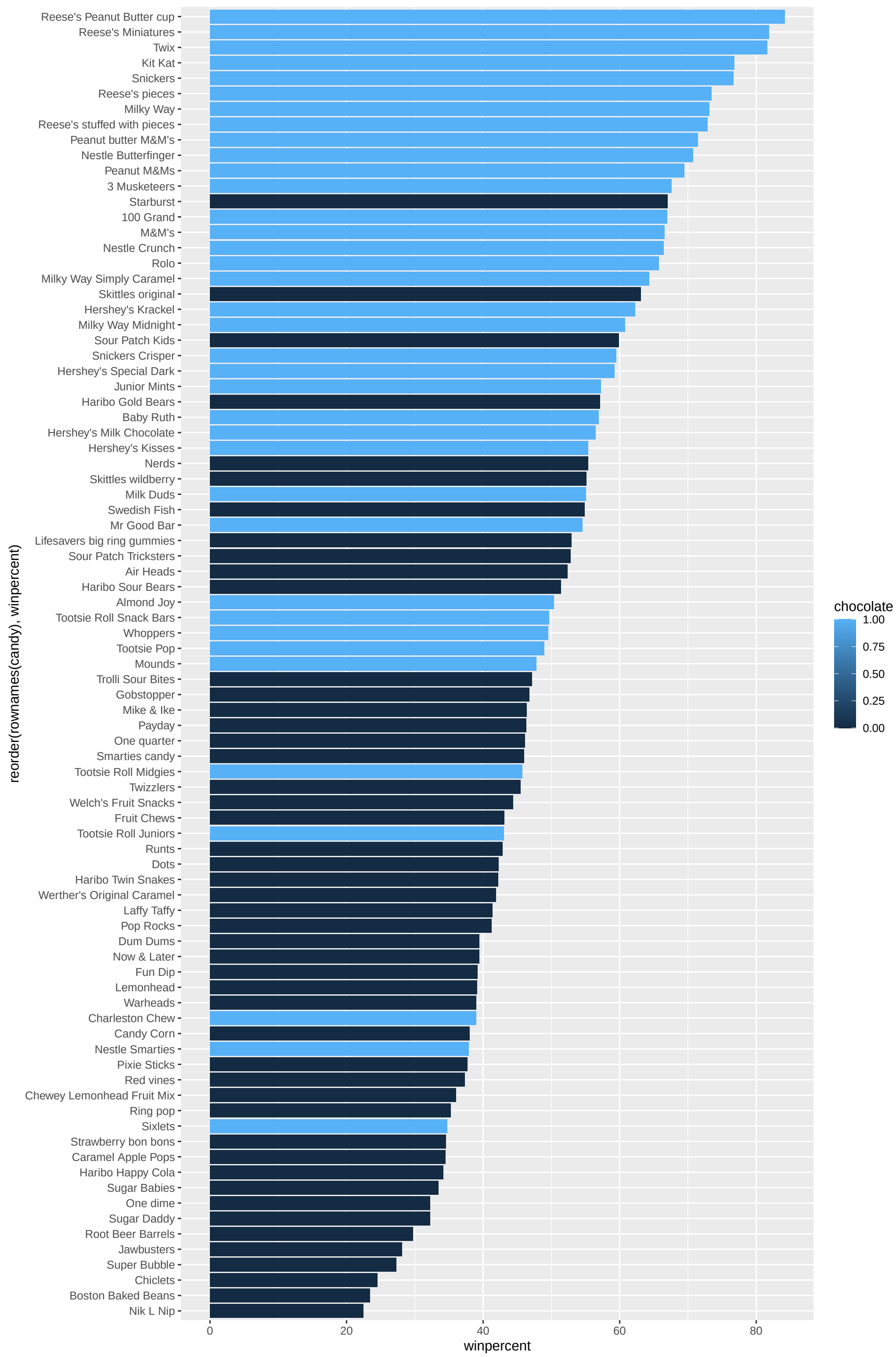
```r
ggplot(candy) +
aes(winpercent, rownames(candy)) +
  geom_col()
```



Whoppers
Werther's Original Caramel
Welch's Fruit Snacks
Warheads
Twizzlers
Twix
Trolli Sour Bites
Tootsie Roll Snack Bars
Tootsie Roll Midgies
Tootsie Roll Juniors
Tootsie Pop
Swedish Fish
Super Bubble
Sugar Daddy
Sugar Babies
Strawberry bon bons
Starburst
Sour Patch Tricksters
Sour Patch Kids
Snickers Crisper
Snickers
Smarties candy
Skittles wildberry
Skittles original
Sixlets
Runts
Root Beer Barrels
Rolo
Ring pop
Reese's stuffed with pieces
Reese's pieces
Reese's Peanut Butter cup
Reese's Miniatures
Red vines
Pop Rocks
Pixie Sticks
Peanut M&Ms
Peanut butter M&M's
Payday
One quarter
One dime
Now & Later
Nik L Nip
Nestle Smarties
Nestle Crunch
Nestle Butterfinger
Nerds
Mr Good Bar
Mounds
Milky Way Simply Caramel
Milky Way Midnight
Milky Way
Milk Duds
Mike & Ike
M&M's
Lifesavers big ring gummies
Lemonhead
Laffy Taffy
Kit Kat
Junior Mints
Jawbusters
Hershey's Special Dark
Hershey's Milk Chocolate
Hershey's Krackel
Hershey's Kisses
Haribo Twin Snakes
Haribo Sour Bears
Haribo Happy Cola
Haribo Gold Bears
Gobstopper
Fun Dip
Fruit Chews
Dum Dums
Dots
Chiclets
Chewey Lemonhead Fruit Mix
Charleston Chew
Caramel Apple Pops
Candy Corn
Boston Baked Beans
Baby Ruth
Almond Joy
Air Heads
3 Musketeers
100 Grand
rownames(candy)
0
20
40
60
80
winpercent

Q16. This is quite ugly, use the reorder() function to get the bars sorted by winpercent?

```
ggplot(candy) +
 aes(x = winpercent, reorder(rownames(candy), winpercent), fill=chocolate) +
  geom_col()
```



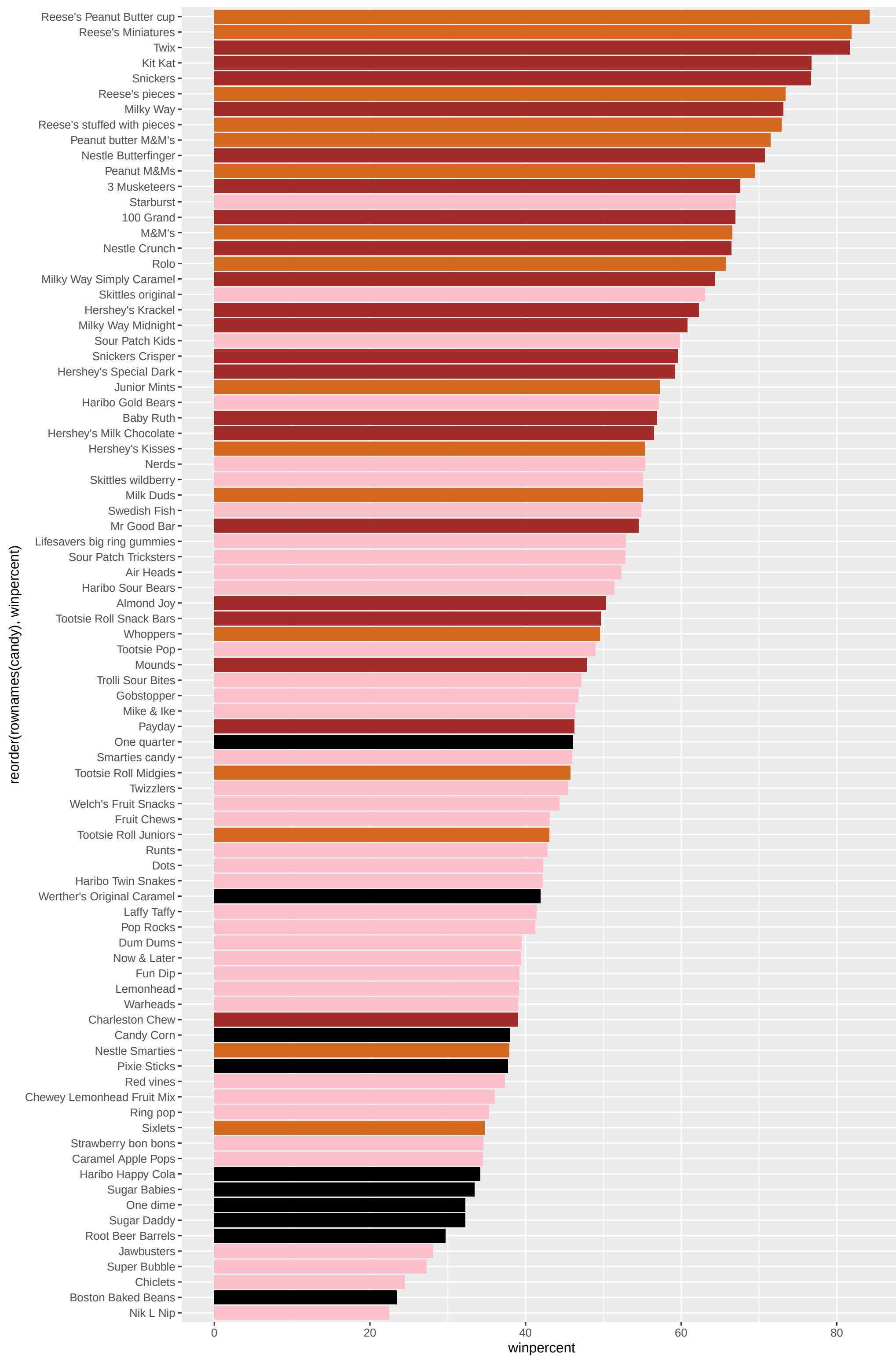
Reese's Peanut Butter cup
Reese's Miniatures
Twix
Kit Kat
Snickers
Reese's pieces
Milky Way
Reese's stuffed with pieces
Peanut butter M&M's
Nestle Butterfinger
Peanut M&Ms
3 Musketeers
Starburst
100 Grand
M&M's
Nestle Crunch
Rolo
Milky Way Simply Caramel
Skittles original
Hershey's Krackel
Milky Way Midnight
Sour Patch Kids
Snickers Crisper
Hershey's Special Dark
Junior Mints
Haribo Gold Bears
Baby Ruth
Hershey's Milk Chocolate
Hershey's Kisses
Nerds
Skittles wildberry
Milk Duds
Swedish Fish
Mr Good Bar
Lifesavers big ring gummies
Sour Patch Tricksters
Air Heads
Haribo Sour Bears
Almond Joy
Tootsie Roll Snack Bars
Whoppers
Tootsie Pop
Mounds
Trolli Sour Bites
Gobstopper
Mike & Ike
Payday
One quarter
Smarties candy
Tootsie Roll Midgies
Twizzlers
Welch's Fruit Snacks
Fruit Chews
Tootsie Roll Juniors
Runts
Dots
Haribo Twin Snakes
Werther's Original Caramel
Laffy Taffy
Pop Rocks
Dum Dums
Now & Later
Fun Dip
Lemonhead
Warheads
Charleston Chew
Candy Corn
Nestle Smarties
Pixie Sticks
Red vines
Chewey Lemonhead Fruit Mix
Ring pop
Sixlets
Strawberry bon bons
Caramel Apple Pops
Haribo Happy Cola
Sugar Babies
One dime
Sugar Daddy
Root Beer Barrels
Jawbusters
Super Bubble
Chiclets
Boston Baked Beans
Nik L Nip
reorder(rownames(candy), winpercent)
0
20
40
60
80
winpercent
chocolate
1.00
0.75
0.50
0.25
0.00

Time to add some useful color

```
mycols <- rep("black", nrow(candy))

## Chocolate candy in chocolate color
mycols [as.logical(candy$chocolate)] <- "chocolate"
## Candy bars in brown
mycols [as.logical(candy$bar)] <- "brown"
## Fruity candy in pink
mycols [as.logical(candy$fruity)] <- "pink"


ggplot(candy) +
aes(x = winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill=mycols)
```


Reese's Peanut Butter cup
Reese's Miniatures
Twix
Kit Kat
Snickers
Reese's pieces
Milky Way
Reese's stuffed with pieces
Peanut butter M&M's
Nestle Butterfinger
Peanut M&Ms
3 Musketeers
Starburst
100 Grand
M&M's
Nestle Crunch
Rolo
Milky Way Simply Caramel
Skittles original
Hershey's Krackel
Milky Way Midnight
Sour Patch Kids
Snickers Crisper
Hershey's Special Dark
Junior Mints
Haribo Gold Bears
Baby Ruth
Hershey's Milk Chocolate
Hershey's Kisses
Nerds
Skittles wildberry
Milk Duds
Swedish Fish
Mr Good Bar
Lifesavers big ring gummies
Sour Patch Tricksters
Air Heads
Haribo Sour Bears
Almond Joy
Tootsie Roll Snack Bars
Whoppers
Tootsie Pop
Mounds
Trolli Sour Bites
Gobstopper
Mike & Ike
Payday
One quarter
Smarties candy
Tootsie Roll Midgies
Twizzlers
Welch's Fruit Snacks
Fruit Chews
Tootsie Roll Juniors
Runts
Dots
Haribo Twin Snakes
Werther's Original Caramel
Laffy Taffy
Pop Rocks
Dum Dums
Now & Later
Fun Dip
Lemonhead
Warheads
Charleston Chew
Candy Corn
Nestle Smarties
Pixie Sticks
Red vines
Chewey Lemonhead Fruit Mix
Ring pop
Sixlets
Strawberry bon bons
Caramel Apple Pops
Haribo Happy Cola
Sugar Babies
One dime
Sugar Daddy
Root Beer Barrels
Jawbusters
Super Bubble
Chiclets
Boston Baked Beans
Nik L Nip
reorder(rownames(candy), winpercent)
0
20
40
60
80
winpercent

Q17. What is the worst ranked chocolate candy?

Sixlets.

```
candy[candy$chocolate == 1,][order(candy[candy$chocolate == 1, ]$winpercent), ][1,]
```

```
        chocolate fruity caramel peanutyalmondy nougat crispedricewafer hard
Sixlets         1      0       0              0      0                0    0
        bar pluribus sugarpercent pricepercent winpercent
Sixlets   0        1         0.22        0.081     34.722
```

Q18. What is the best ranked fruity candy?

Starburst.

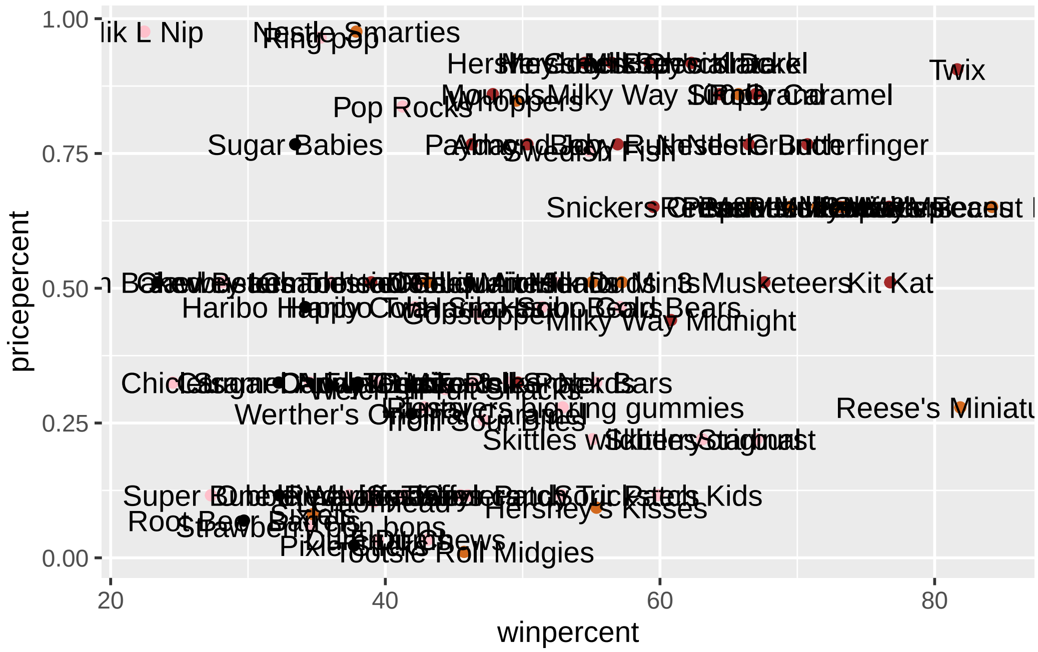
```
candy[candy$fruity == 1, ][order(-candy[candy$fruity == 1, ]$winpercent), ][1, ]
```

```
          chocolate fruity caramel peanutyalmondy nougat crispedricewafer hard
Starburst         0      1       0              0      0                0    0
          bar pluribus sugarpercent pricepercent winpercent
Starburst   0        1        0.151         0.22   67.03763
```

Taking a look at pricepercent

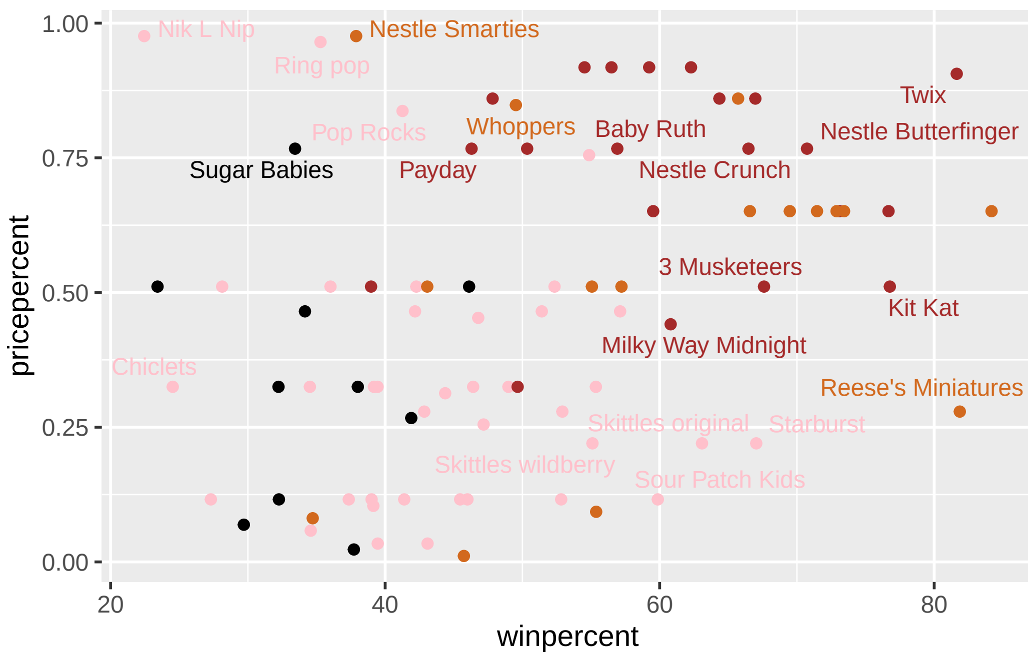
Make a plot of winpercent vs pricepercent

```
ggplot(candy) +
  aes(winpercent, pricepercent,
      label=rownames(candy)) +
  geom_point(color = mycols) +
  geom_text()
```



We can fix the label text overplotting with an add-on package called **ggrepel** and it's `geom_text_repel()`

```
library(ggrepel)
ggplot(candy) +
  aes(winpercent, pricepercent,
      label = rownames(candy)) +
  geom_point(col = mycols) +
  geom_text_repel(col = mycols, size = 3.3, max.overlaps = 5)
```



> Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Tootsie Roll Midgies.

```
candy[order(candy$pricepercent, -candy$winpercent), ][1, ]
```

```
                     chocolate fruity caramel peanutyalmondy nougat
Tootsie Roll Midgies         1      0       0              0      0
                     crispedricewafer hard bar pluribus sugarpercent
Tootsie Roll Midgies                0    0   0        1        0.174
                     pricepercent winpercent
Tootsie Roll Midgies        0.011   45.73675
```

> Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

The top 5 most expensive candy types are Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, and Hershey's Milk Chocolate. The least popular amongst these is Nik L Nip.

```
ord <- order(candy$pricepercent, decreasing = TRUE)
top5 <- candy[ord, ][1:5, ]
top5
```

```
                          chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                         0      1       0              0      0
Nestle Smarties                   1      0       0              0      0
Ring pop                          0      1       0              0      0
Hershey's Krackel                 1      0       0              0      0
Hershey's Milk Chocolate          1      0       0              0      0
                          crispedricewafer hard bar pluribus sugarpercent
Nik L Nip                                0    0   0        1        0.197
Nestle Smarties                          0    0   0        1        0.267
Ring pop                                 0    1   0        0        0.732
Hershey's Krackel                        1    0   1        0        0.430
Hershey's Milk Chocolate                 0    0   1        0        0.430
                          pricepercent winpercent
Nik L Nip                        0.976   22.44534
Nestle Smarties                  0.976   37.88719
Ring pop                         0.965   35.29076
Hershey's Krackel                0.918   62.28448
Hershey's Milk Chocolate         0.918   56.49050
```

```
top5[which.min(top5$winpercent), ]
```

```
          chocolate fruity caramel peanutyalmondy nougat crispedricewafer hard
Nik L Nip         0      1       0              0      0                0    0
          bar pluribus sugarpercent pricepercent winpercent
Nik L Nip   0        1        0.197        0.976   22.44534
```

Exploring the correlation structure

We can calculate the pair-wise correlation of all our columns

```
cij <- cor(candy)
cij
```

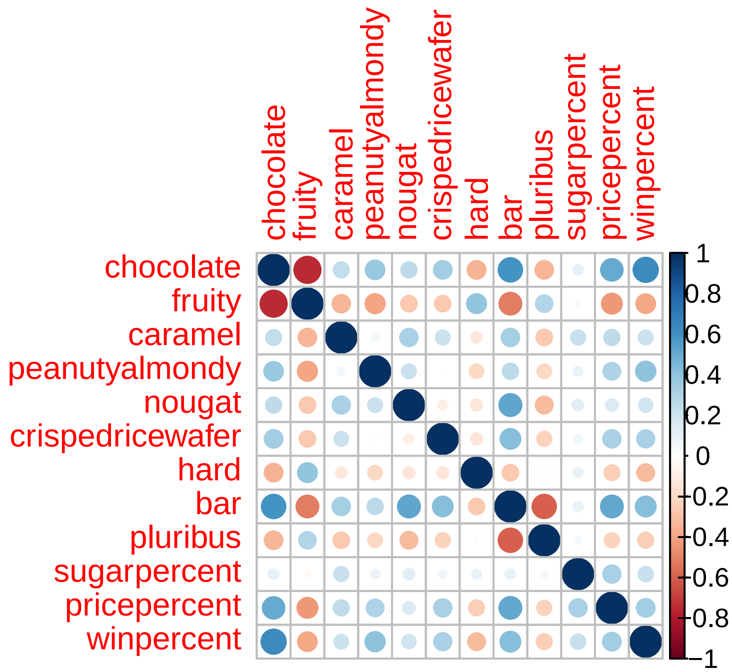
```
                  chocolate      fruity     caramel peanutyalmondy      nougat
chocolate         1.0000000 -0.74172106  0.24987535     0.37782357  0.25489183
fruity           -0.7417211  1.00000000 -0.33548538    -0.39928014 -0.26936712
caramel           0.2498753 -0.33548538  1.00000000     0.05935614  0.32849280
peanutyalmondy    0.3778236 -0.39928014  0.05935614     1.00000000  0.21311310
nougat            0.2548918 -0.26936712  0.32849280     0.21311310  1.00000000
crispedricewafer  0.3412098 -0.26936712  0.21311310    -0.01764631 -0.08974359
```

```
hard             -0.3441769  0.39067750 -0.12235513    -0.20555661 -0.13867505
bar               0.5974211 -0.51506558  0.33396002     0.26041960  0.52297636
pluribus         -0.3396752  0.29972522 -0.26958501    -0.20610932 -0.31033884
sugarpercent      0.1041691 -0.03439296  0.22193335     0.08788927  0.12308135
pricepercent      0.5046754 -0.43096853  0.25432709     0.30915323  0.15319643
winpercent        0.6365167 -0.38093814  0.21341630     0.40619220  0.19937530
                 crispedricewafer        hard         bar    pluribus
chocolate              0.34120978 -0.34417691  0.59742114 -0.33967519
fruity                -0.26936712  0.39067750 -0.51506558  0.29972522
caramel                0.21311310 -0.12235513  0.33396002 -0.26958501
peanutyalmondy        -0.01764631 -0.20555661  0.26041960 -0.20610932
nougat                -0.08974359 -0.13867505  0.52297636 -0.31033884
crispedricewafer       1.00000000 -0.13867505  0.42375093 -0.22469338
hard                  -0.13867505  1.00000000 -0.26516504  0.01453172
bar                    0.42375093 -0.26516504  1.00000000 -0.59340892
pluribus              -0.22469338  0.01453172 -0.59340892  1.00000000
sugarpercent           0.06994969  0.09180975  0.09998516  0.04552282
pricepercent           0.32826539 -0.24436534  0.51840654 -0.22079363
winpercent             0.32467965 -0.31038158  0.42992933 -0.24744787
                 sugarpercent pricepercent winpercent
chocolate          0.10416906    0.5046754  0.6365167
fruity            -0.03439296   -0.4309685 -0.3809381
caramel            0.22193335    0.2543271  0.2134163
peanutyalmondy     0.08788927    0.3091532  0.4061922
nougat             0.12308135    0.1531964  0.1993753
crispedricewafer   0.06994969    0.3282654  0.3246797
hard               0.09180975   -0.2443653 -0.3103816
bar                0.09998516    0.5184065  0.4299293
pluribus           0.04552282   -0.2207936 -0.2474479
sugarpercent       1.00000000    0.3297064  0.2291507
pricepercent       0.32970639    1.0000000  0.3453254
winpercent         0.22915066    0.3453254  1.0000000
```

```r
library(corrplot)
```

```
corrplot 0.95 loaded
```

```r
corrplot(cij)
```



> Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Chocolate and fruity are strongly anti-correlated.

> Q23. Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

Chocolate and fruity are expected to have the largest contributions to PC1. They show strong opposing patterns and represent the main separation in the dataset.

Principal Component Analysis

In this case we want to be sure to set `scale=TRUE` argument for `prcomp` because we have one variable, `winpercent` that is on a very different scale than all others and would otherwise dominate our PCA results.

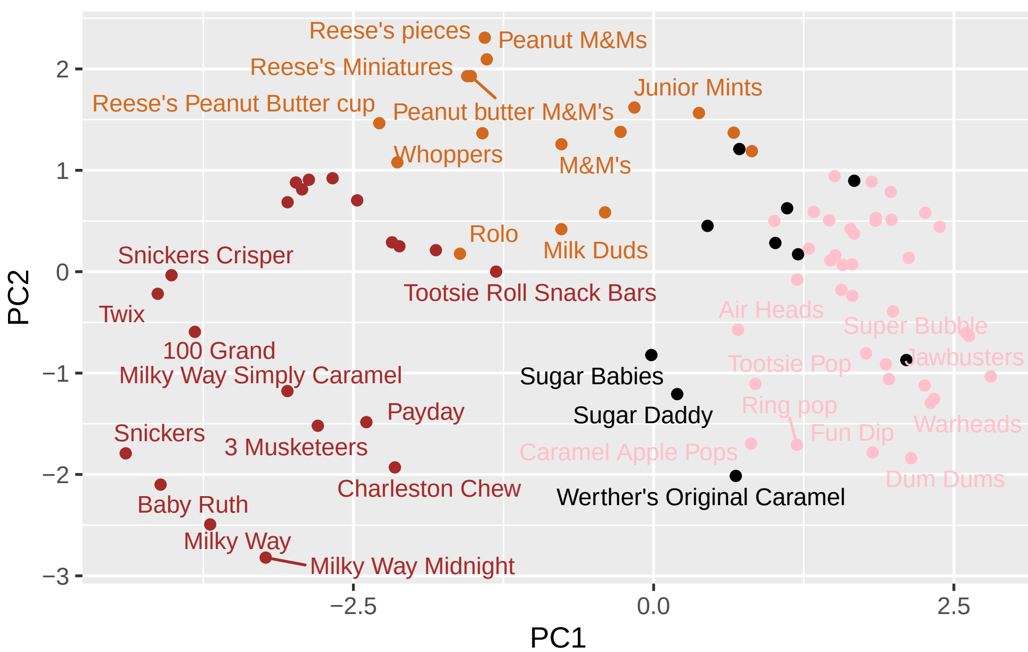
```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
```

```
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

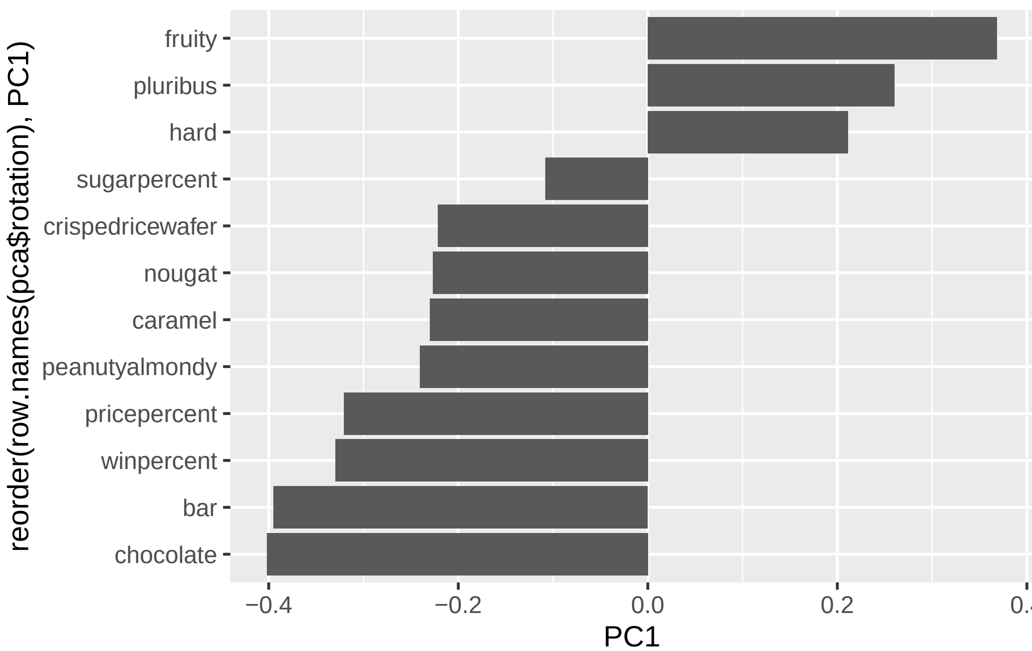
First major result figure is the "score plot" of PC1 vs PC2 - how the different candy are related to each other on our new PC axis:

```r
ggplot(pca$x) +
  aes(PC1,PC2, label=row.names(pca$x)) +
  geom_point(col=mycols) +
  geom_text_repel(size=3.3, col=mycols, max.overlaps = 7)
```



The second major results figure from PCA is the so-called "loadings plot".

```r
ggplot(pca$rotation) +
  aes(PC1, reorder(row.names(pca$rotation), PC1)) +
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

(SEE CODE ABOVE). The variables fruity, pluribus, and hard have the strongest positive loadings on PC1. This makes sense because these features are associated with similar types of candies. This relationship was observed in the correlation plot, where these variables showed positive correlations with each other and negative correlation with chocolate.

Summary

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a "winning" candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A "winning" candy in this dataset tends to be chocolate-based, often a bar, and frequently includes features like caramel, peanuts, or crisped rice, rather than being fruity or hard candy.